NetSentinel

AI copilot for NOCs.

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An AI copilot for telecom network operations

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Open source on GitHub
github.com/ajay206/NetSentinel

The problem

One fibre cut, 40+ alarms

When one fibre is cut, the NOC sees 40+ alarms at once. Most of them are symptoms of a single root cause.

The NOC needs the cause and the fix, not more alarms.

Alarm storms

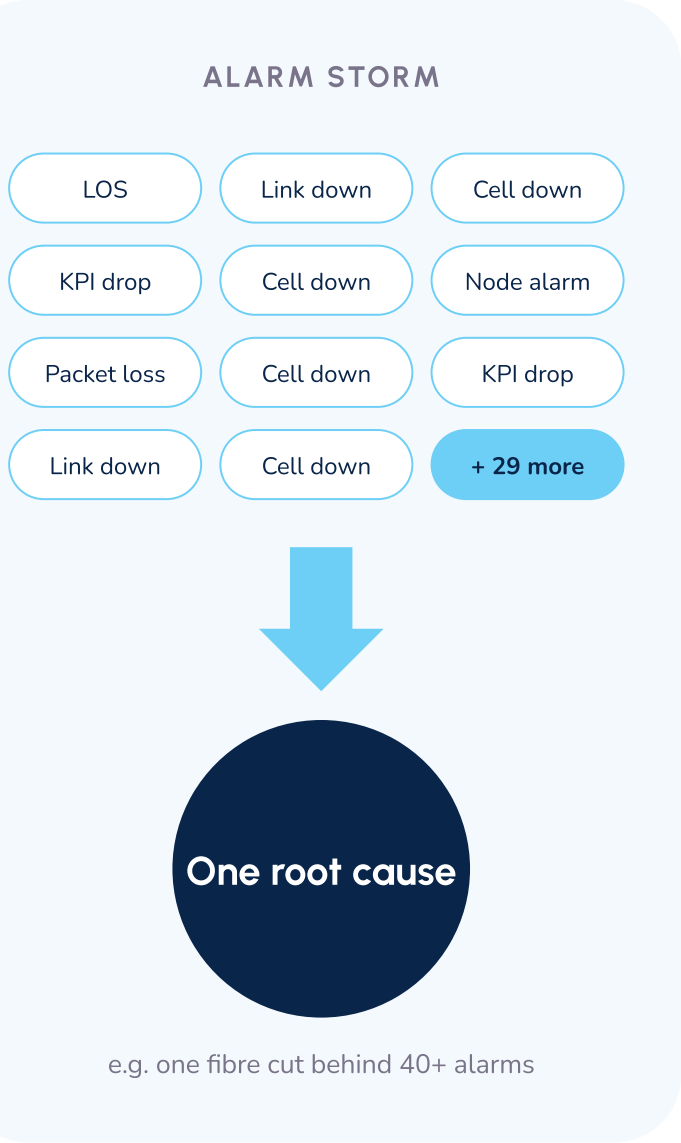
The real fault is buried under symptoms.

Slow RCA

Root cause is traced by hand.

Siloed tools

Topology, KPIs and tickets live apart.

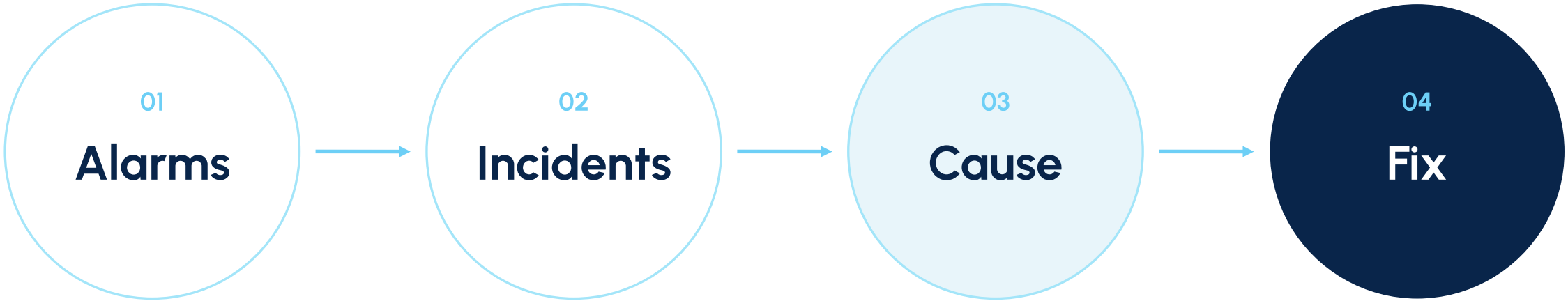


e.g. one fibre cut behind 40+ alarms

What it does

Alarms in, fixes out

One dashboard takes the NOC from raw alarms to an approved fix.



Telemetry and alarms stream in from the network

Related alarms are grouped into one incident

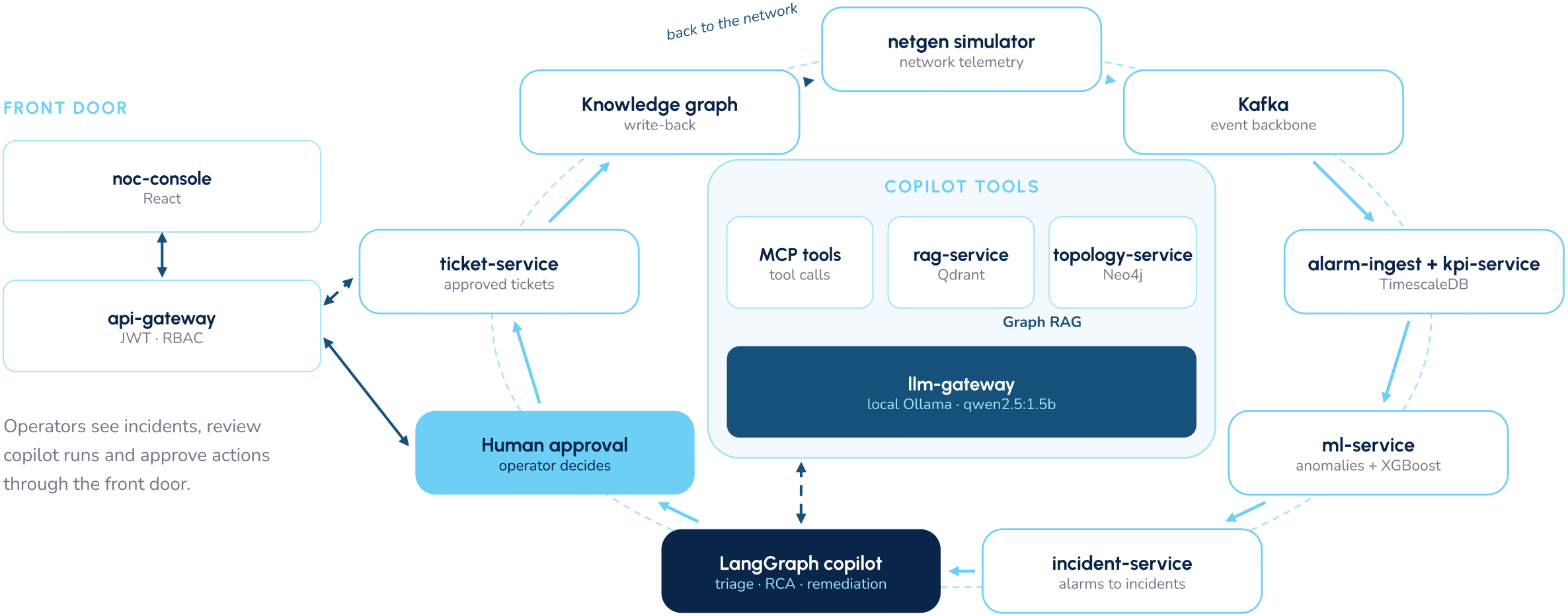
ML and the topology graph find the root cause

The copilot drafts the fix; an operator approves it

All in one screen: the NOC console

Architecture

A closed loop, alarm to fix



Operators see incidents, review copilot runs and approve actions through the front door.

Data pipeline

Telemetry to clean KPIs

Every event flows through Kafka, so each service can be replaced or scaled on its own.

01

Simulate

netgen

A synthetic network simulator emits telemetry and alarms

02

Stream

Kafka

The event backbone between every service

03

Enrich

alarm-ingest

Raw alarms get network context before they are stored

04

Store

kpi-service

KPI time series land in TimescaleDB

100% synthetic data: no operator or customer data.

Spot it, then name it

ANOMALY DETECTION

z-score + CUSUM

Statistical detectors flag sudden spikes and slow drift in KPIs.

FAULT CLASSIFIER

XGBoost

Names the likely fault type behind an anomaly.

EXPLAIN & TRACK

SHAP + MLflow

SHAP shows which signals drove each prediction. MLflow tracks the models.

Runs in ml-service, on the KPIs stored in TimescaleDB.

Topology graph

Blast radius in ~5 ms

NEO4J IMPACT QUERY

"If link LNK-012 fails, which cells lose service?"

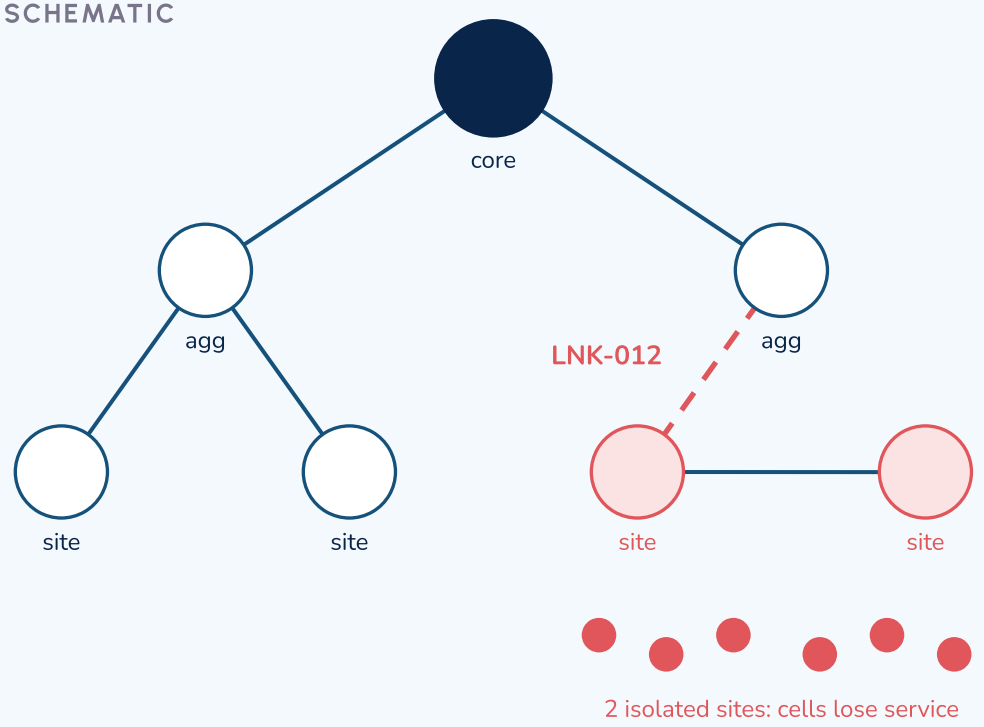
~5 ms
to answer

12
cells lost

2
sites isolated

Real run on the synthetic network. The same graph powers RCA: many alarms, one upstream cause.

SCHEMATIC



Graph RAG, measured

1

Entity linking

Find the network elements a question names

2

10 fixed graph queries

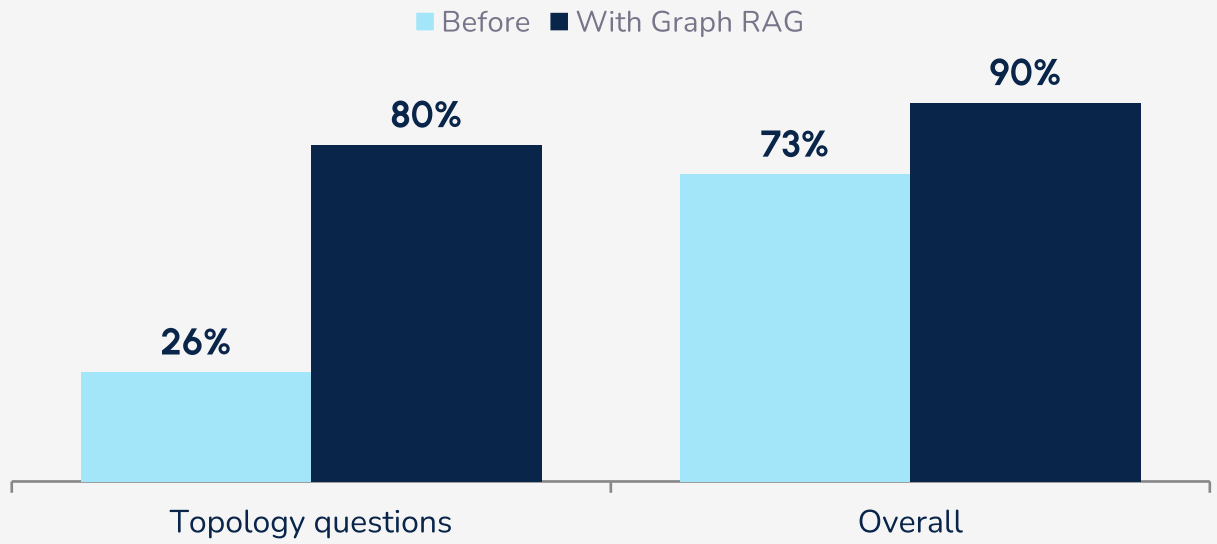
Read-only, no free-form query writing

3

Reranker

Puts the best evidence first

HIT@1 ON A 150-QUESTION BENCHMARK



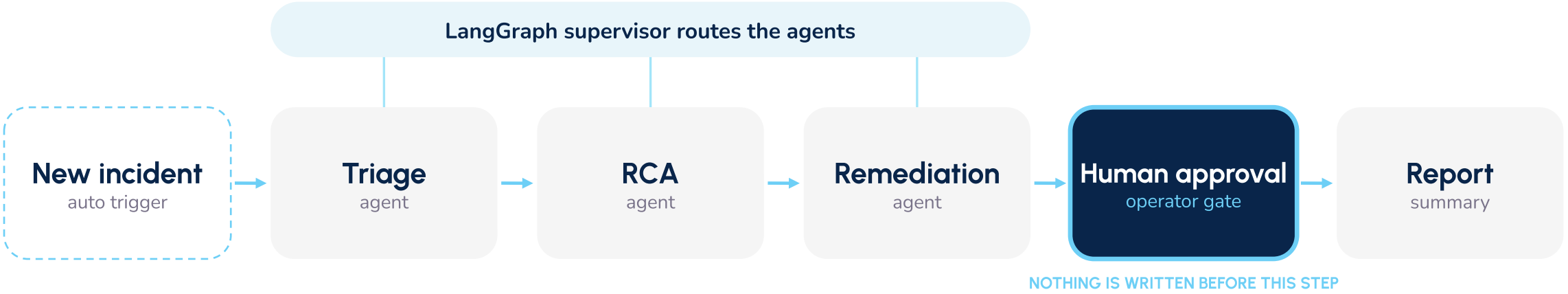
p95 ~2.5 ms
retrieval latency

150 questions
benchmark

Synthetic
network data

Multi-agent copilot

Agents propose, humans approve



Approval first

Nothing is written until an operator approves.

Approval queue

Ticket drafts wait in the queue for review.

Traced

Every step is visible in the console.

Automatic

Runs start when a new incident opens.

LLM gateway & guardrails

One door to the model

Circuit breakers

Fail fast when a model backend is unhealthy

Prompt-injection guard

Screens inputs before they reach the model

llm-gateway

48/48

red-team cases pass

The only service that calls a model. Local model first.

PII redaction

Strips personal data from prompts

Per-caller rate limits

Each calling service gets its own limit

CI check

The build fails if any service calls an LLM directly.

Security

Locked down
and post-quantum

Hybrid post-quantum TLS on the wire, signed models on disk, and a tamper-evident trail of every action.

01

Access & audit

- RBAC: viewer, operator, admin
- bcrypt-hashed users
- Lockout and change password
- Hash-chained, tamper-evident audit log

02

Post-quantum

- Hybrid TLS X25519MLKEM768
- Classical-only clients refused
- ML-DSA-65 signed models
- A tampered model refuses to load

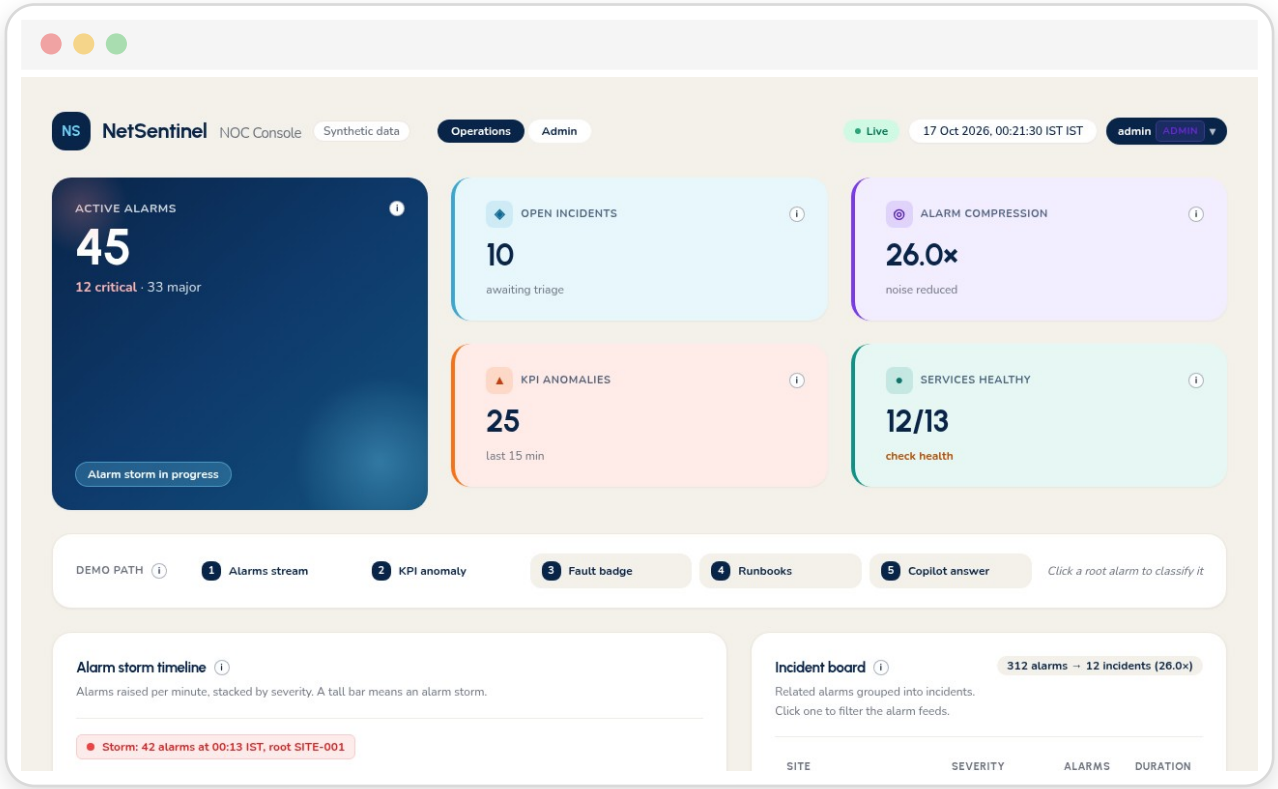
HANDSHAKE OVERHEAD

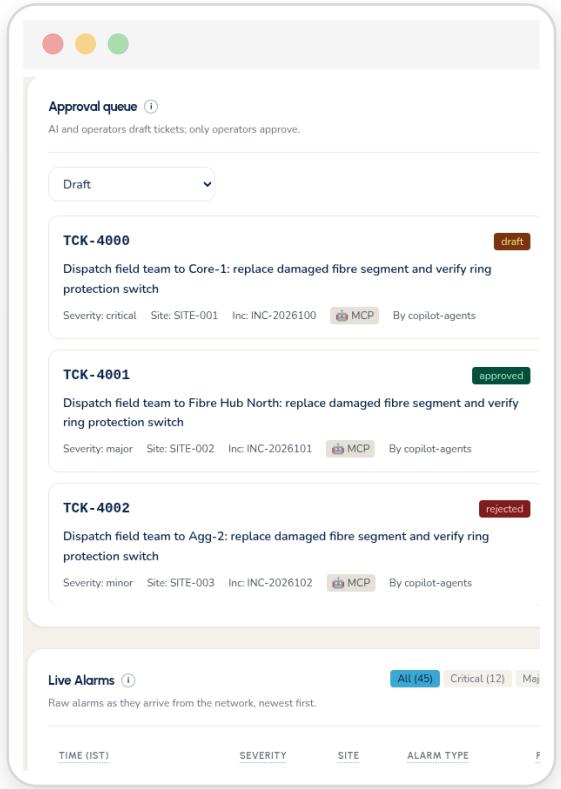
+0.12 ms

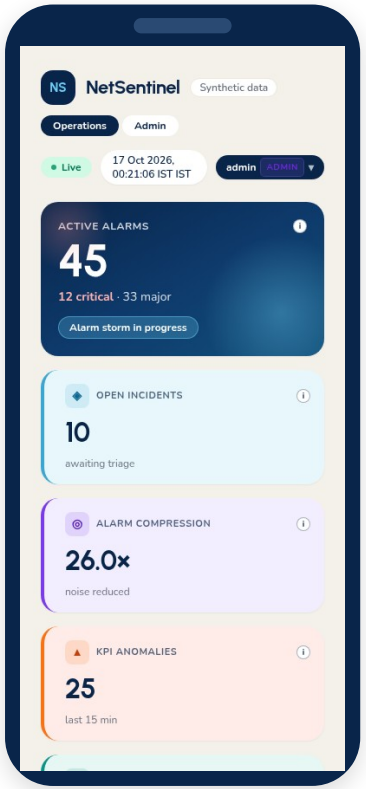
about, for hybrid post-quantum TLS. Built on OpenSSL 3.5.

One bento console, phone to desktop

The NOC console: bento KPI tiles, an approval queue and live alarms, responsive from phone to desktop.







Engineering quality

Built like production

600+

pytest tests

175+

Vitest tests

1

command: make demo

- GitHub Actions CI: merge only on green
- Docker only: Mac, Linux and Windows
- Junior-developer handbook
- ADRs for key design decisions
- OpenAPI contracts for every API
- 100% synthetic data

Tech stack

Built on open tools

One Docker-only make demo brings the whole stack up.

- Python / FastAPI
- Kafka
- TimescaleDB
- Neo4j
- Qdrant
- LangGraph
- MCP
- Ollama
- React / Vite
- Docker
- OpenSSL 3.5 PQC

Star the repo

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Open to telecom AI roles. Let's talk.